Production-Ready E-Commerce Platform on Azure

Portfolio Project Architecture & Implementation Roadmap

This single project is designed to cover nearly every core Azure service — identity, networking, Kubernetes, data, messaging, CI/CD, observability, and security — while staying within a $200 Azure credit, provided resources are shut down when not in use.

Estimated timeline	6–8 weeks (portfolio pace)
Budget target	Within a $200 Azure credit
Core stack	Terraform · AKS · Azure DevOps · Key Vault · Service Bus
Project name	Azure Enterprise DevOps Platform

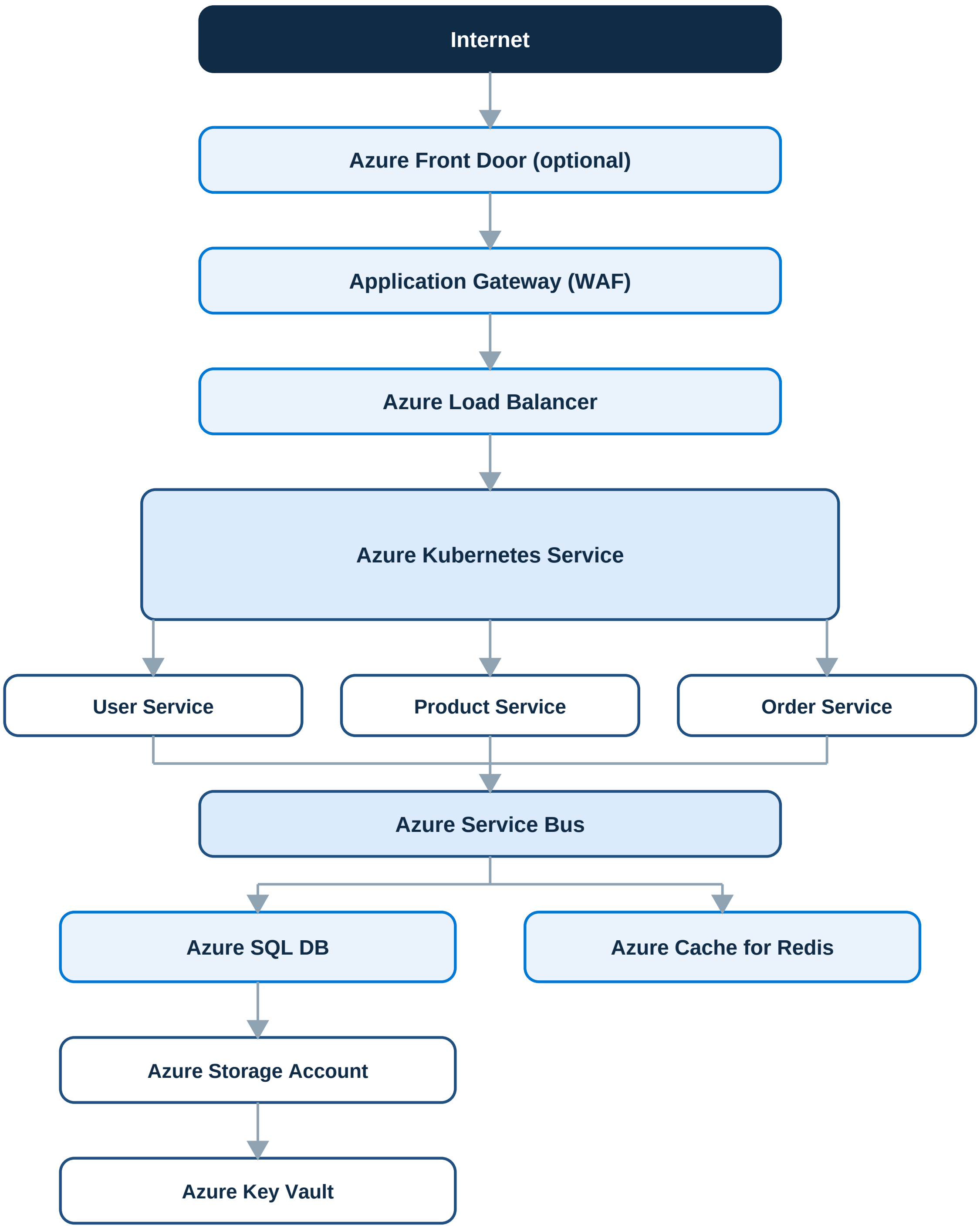
Contents

Overview

This single project can cover nearly every core Azure service while staying within a $200 credit, as long as resources are shut down when they are not in use. It is structured as a phased build: identity and networking first, then infrastructure as code, containers, data, messaging, CI/CD, observability, and security — in that order, so that each phase builds on infrastructure the previous phase already put in place.

Final Architecture

Traffic enters through Application Gateway (acting as a WAF) and an Azure Load Balancer in front of an AKS cluster running three microservices. The services communicate asynchronously through Azure Service Bus and persist data to Azure SQL Database and Azure Cache for Redis, with Azure Storage and Key Vault supporting file storage and secrets.



Cross-Cutting Concerns

Monitoring

- Azure Monitor
- Application Insights
- Log Analytics

Identity

- Microsoft Entra ID
- Managed Identity
- Azure RBAC

CI/CD

- Azure DevOps Repos
- Azure Pipelines
- Azure Container Registry
- Terraform

Services You'll Learn

Area	Azure Service
IAM	Microsoft Entra ID
RBAC	Azure RBAC
Secrets	Azure Key Vault
Networking	VNet, Subnets
Security	NSG, WAF, Defender (trial)
Container Registry	Azure Container Registry
Kubernetes	AKS
Databases	Azure SQL, Cosmos DB (optional)
Object Storage	Storage Account
Messaging	Service Bus
Caching	Azure Redis
Monitoring	Azure Monitor
Logs	Log Analytics
Tracing	Application Insights
CI/CD	Azure DevOps Pipelines
IaC	Terraform

Area	Azure Service
DNS	Azure DNS
Load Balancer	Azure Load Balancer
Reverse Proxy	Application Gateway

Suggested Folder Structure

```
azure-devops-project/

terraform/
    networking/
    aks/
    sql/
    keyvault/
    monitoring/

kubernetes/
    ingress/
    deployments/
    services/
    hpa/

pipelines/
    ci.yml
    cd.yml

services/
    user-service
    product-service
    order-service

helm/
docs/
scripts/
```

Implementation Roadmap

Thirteen phases, each building on Azure infrastructure the previous phase already established. Phases 1–12 form the core build; Phase 13 is an optional GitOps upgrade.

Phase 1 — Identity

Create

- Azure Subscription
- Resource Group
- Microsoft Entra ID
- Azure RBAC
- Managed Identity
- Service Principals

Skills

- IAM
- Authentication
- Authorization

Phase 2 — Networking

Create

- VNet
- Multiple Subnets
- NSGs
- Route Tables
- Public IP
- NAT Gateway

Learn

- CIDR
- Private IP
- Public IP
- Service Endpoints
- Private Endpoints

Phase 3 — Infrastructure as Code

Provision everything using

- Terraform

Structure — Modules

- network
- aks
- sql
- monitoring
- keyvault
- acr

No portal clicks except the initial subscription setup.

Phase 4 — Azure Container Registry

Build Docker images

- User Service
- Product Service
- Order Service

Push

- Azure Container Registry

Phase 5 — AKS

Deploy

- AKS Cluster
- Node Pool
- Ingress Controller
- HPA
- ConfigMaps
- Secrets
- Persistent Volumes

Learn

- Pods
- Deployments
- Services
- Ingress
- Autoscaling

Microsoft's AKS secure baseline architecture is an excellent reference for how production clusters are designed.

Phase 6 — Databases

Deploy

- Azure SQL

Tables

- Users
- Orders
- Products

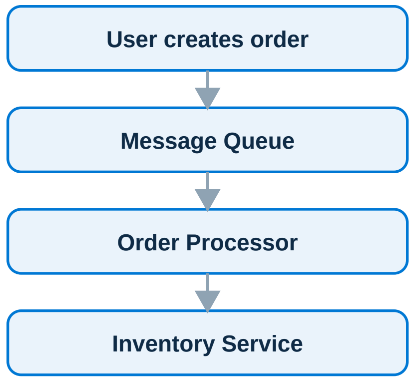
Later

- Add Cosmos DB

Phase 7 — Messaging

Use

- Azure Service Bus



Phase 8 — Storage

Upload

- Images
- Invoices
- Logs

Using

- Azure Storage Account

Phase 9 — Secrets

Store

- Database Password
- JWT Secret
- Storage Key
- API Keys

Inside

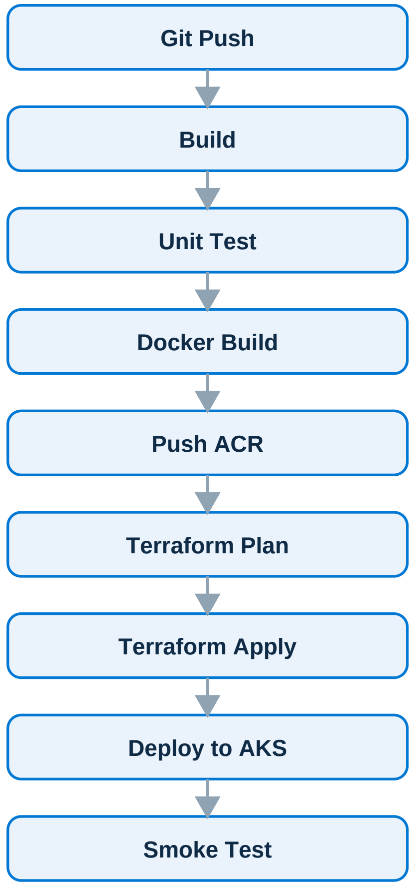
- Azure Key Vault

AKS accesses secrets through Managed Identity rather than storing credentials in code.

Phase 10 — CI/CD

Platform

- Azure DevOps



Phase 11 — Monitoring

Enable

- Azure Monitor
- Log Analytics
- Application Insights

Collect

- CPU
- Memory
- Requests
- Exceptions
- Container Logs

Create

- Alerts

Phase 12 — Security

Implement

- Network Policies
- Azure Policy
- RBAC
- Key Vault
- Private Endpoints
- WAF
- Defender for Cloud (trial)

The Azure DevSecOps guidance for AKS shows how to integrate security throughout the CI/CD lifecycle rather than adding it only at deployment time.

Optional Phase 13 — GitOps

Instead of deploying directly, use

- FluxCD
- ArgoCD

Git becomes the source of truth.

Skills You'll Gain

By the end, you'll have practical, hands-on experience with:

- Azure Networking
- Azure IAM
- Azure RBAC
- Azure Kubernetes Service
- Azure DevOps
- Azure Monitor
- Azure SQL
- Azure Storage
- Azure Key Vault
- Azure Container Registry
- Azure Service Bus
- Terraform
- Docker
- Kubernetes
- GitOps
- DevSecOps
- Observability

Estimated Monthly Cost (Lab Use)

If resources only run while actively practicing, and are deleted afterward:

Resource	Estimated Monthly Cost
AKS (small cluster)	~$40–70
Azure SQL (Basic)	~$5–10

Resource	Estimated Monthly Cost
ACR (Basic)	~$5
Storage	<$2
Log Analytics	Depends on ingestion (keep it low)
Key Vault	Negligible
Service Bus (Basic)	Low cost
Azure DevOps	Free tier is sufficient for learning

Staying under the $200 Azure credit is realistic if resources are stopped or deleted after each lab session and AKS is not left running 24/7.

What I'd Build

Treat this as a 6–8 week portfolio project called **Azure Enterprise DevOps Platform**. Every component would be provisioned with Terraform, deployed through Azure DevOps Pipelines, run on AKS, secured with Microsoft Entra ID and Key Vault, monitored with Azure Monitor and Application Insights, and documented with architecture diagrams and a README. That combination produces a project that can be discussed confidently in interviews, because it mirrors how many production Azure environments are actually built. Microsoft's AKS baseline reference architecture and the Landing Zone Accelerator are useful references at each stage.